25901385 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMAT201 - FUNDAMENTALS OF ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. Exhibit a bijection between $\mathbb{N}$ and the set of all odd integers. [U] / [CO1]
2. Determine the set A of all real numbers x such that $2x + 3 \leq 6$. [A] / [CO2]
3. Give an example of a set that is bounded above, not below. [U] / [CO3]
4. Define a half-closed interval. [K] / [CO4]
5. Show that $\overline{e^{i\theta}} = e^{-i\theta}$. [U] / [CO5]
6. Give an example of a set that is neither open nor closed in C. [U] / [CO6]
7. Evaluate $Log\,(-1)$ [A] / [CO8]
8. Find the value of z = x+iy such that $e^z = 1 + i$. [An] / [CO8]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Show that $1 < 2 < 3 < 4 < \ldots$ [A] / [CO2]

10. Show that for all $a, b \in \mathbb{R}, \quad \min\{a, b\} = \frac{1}{2}(a + b - |a - b|)$. [A] / [CO2]

11. Let S be a nonempty subset of $\mathbb{R}$ that is bounded above, and let a be any number in $\mathbb{R}$. Define the set $a + S = \{a + s : s \in S\}$. Prove that $\sup(a + S) = a + \sup S$. [U] / [CO3]

12. Write the rational number represented by the periodic decimal 7.31414...14... [An] / [CO4]

13. Solve for the real and imaginary parts of the function (a) $f(z) = iz + 2$ and (b) $f(z) = x^2y + i(xy^2)$. In each part, check whether the Cauchy–Riemann equations are satisfied or not. [A] / [CO6]

14. Determine whether the function $u(x, y) = (x^2 - y^2)^2$ is harmonic. [An] / [CO7]

15. Prove that $|\cos z| \geq |\cos x|$ [U] / [CO8]

16. Prove that $\cosh^2 z - \sinh^2 z = 1$. [A] / [CO8]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. a) Give an example of a countable collection of finite sets whose union is not finite. [U] / [CO1]
    b) Show that the union of two disjoint denumerable sets is denumerable.

18. a) Let S be a nonempty subset of $\mathbb{R}$ that is bounded below. Prove that $infS = -supf\{-s : s \in S\}$. [An] / [CO3]
    b) Show that if a nonempty set S has a finite number of elements, then the $supS \in S$.

19. a) State and prove the density theorem. [A] / [CO3]
    b) Can the same argument be said about irrationals? Justify your answer.

20. Evaluate the following [U] / [CO5]
    a) $Re\frac{(2-3i)^2}{(2+3i)}$
    b) $Im\ z^3$
    c) $Im\frac{z}{\bar{z}}$.

21. a) Find all the nth roots of z where $n = 1, 2, 3, \ldots$. [E] / [CO8]
    b) Prove $\log(z^{\frac{1}{n}}) = \frac{1}{n}\ \ln r + i(\frac{\Theta + 2(pn+k)\pi}{n})$.

22. a) Derive the logarithmic expression of $\sin^{-1} z$. [An] / [CO8]
    b) Find $\frac{d}{dx}(\sin^{-1} z)$.

[10x3 = 30]